1 Functions f and g are defined on the set of real numbers by

- $f(x) = x^2 + 3$
- $g(x) = x + 4$

(a) Find expressions for:

(i) $f(g(x))$;

(ii) $g(f(x))$. 3

Generic Scheme

1 (a)

- $\bullet^1$ ic start composite process
- $\bullet^2$ ic correct substitution into expression
- $\bullet^3$ ic complete second composite

Illustrative Scheme

- $\bullet^1$ e.g. $f(x+4)$ **stated, or implied by** $\bullet^2$
- $\bullet^2$ $(x+4)^2 + 3$
- $\bullet^3$ $x^2 + 3 + 4$

3

Notes

1. Candidates must clearly identify which of their answers are $f(g(x))$ and $g(f(x))$; the minimum evidence for this could be as little as using (i) and (ii) as labels.
2. Candidates who interpret the composite functions as either $f(x) \times g(x)$ or $f(x) + g(x)$, do not gain any marks.

Regularly occurring responses

Response 1 : The first two marks are for **either** $f(g(x))$ **or** $g(f(x))$ correct. The third mark is for the other composite function.

Candidate A

$f(g(x))$
$= (x+4)^2 + 3$ ✓ $\bullet^1$ ✓ $\bullet^2$

$g(f(x))$
$= x^2 + 12$ ✗ $\bullet^3$

2 marks out of 3

Candidate B

$f(g(x))$
$= (x+7)^2$ ✗ $\bullet^3$

$g(f(x))$
$= x^2 + 7$ ✓ $\bullet^1$ ✓ $\bullet^2$

2 marks out of 3

Response 2 : Interpreting $f(g(x))$ as $g(f(x))$ and vice versa. A maximum of 2 marks are available.

Candidate C

$f(g(x))$
$= x^2 + 7$ ✗ $\bullet^1$ ✓ $\bullet^2$

$g(f(x))$
$= (x+4)^2 + 3$ ✓ $\bullet^3$

2 marks out of 3

Candidate D

$f(g(x))$
$= x^2 + 7$ ✗ $\bullet^1$ ✓ $\bullet^2$

1 mark out of 3

Response 3 : Identifying $f(g(x))$ and $g(f(x))$

Candidate E

$(x+4)^2 + 3$ ✗ $\bullet^1$ ✓ $\bullet^2$
$x^2 + 7$ ✓ $\bullet^3$

2 marks out of 3

Candidate F

$x^2 + 7$ ✗ $\bullet^1$ ✗ $\bullet^2$
$(x+4)^2 + 3$ ✓ $\bullet^3$

1 mark out of 3

Candidate G

$x^2 + 7$ ONLY
or $(x+4)^2 + 3$ ONLY

0 marks out of 3

Candidate H

(i) $(x+4)^2 + 3$ ✓ $\bullet^1$ ✓ $\bullet^2$
(ii) $x^2 + 7$ ✓ $\bullet^3$

3 marks out of 3

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

1	(b)	Show that $f\big(g(x)\big)+g\big(f(x)\big)=0$ has no real roots.	**3**

Generic Scheme

1 (b)

Method 1 : Discriminant

- $\bullet^4$ pd obtain a quadratic expression
- $\bullet^5$ ss know to and use discriminant
- $\bullet^6$ ic interpret result

Method 2 : Quadratic Formula

- $\bullet^4$ pd obtain a quadratic expression
- $\bullet^5$ ss know to and use quadratic formula
- $\bullet^6$ ic interpret result

Illustrative Scheme

Method 1 : Discriminant

- $\bullet^4$ $2x^2+8x+26$
- $\bullet^5$ $8^2-4\times2\times26$ or $4^2-4\times1\times13$ **stated, or implied by** $\bullet^6$
- $\bullet^6$ $-144<0$ or $-36<0$ so no real roots

Method 2 : Quadratic Formula

- $\bullet^4$ $2x^2+8x+26$
- $\bullet^5$ $\dfrac{-8\pm\sqrt{8^2-4\times2\times26}}{2\times2}$ **stated, or implied by** $\bullet^6$
- $\bullet^6$ $\sqrt{-144}$ not possible so no real roots

3

Notes

3. Candidates who use $f(x)\times g(x)$ can gain no marks in (b) as a cubic will be obtained.
4. Candidates who use $f(x)+g(x)$ do not gain $\bullet^4$ (eased) but $\bullet^5$ and $\bullet^6$ are available as follow through marks.
5. In method 1, any other formula masquerading as a discriminant cannot gain $\bullet^5$ and $\bullet^6$.
6. $\bullet^4$, $\bullet^5$ and $\bullet^6$ are only available if $f(g(x))+g(f(x))$ simplifies to a quadratic expression of the form ax^2+bx+c, with b and c both non-zero.
7. $\bullet^6$ is only available for a numerical value, calculated correctly from the candidate's response at $\bullet^4$, and leading to no real roots.
8. Do not accept for $\bullet^6$:
   - 'no roots' in lieu of 'no real roots'
   - 'maths error' or 'ma error'.
9. Candidates who use the word derivative instead of discriminant should not be penalised.

Regularly occurring responses

Response 4 : Candidates who do not simplify the value of their discriminant

Candidate I

$8^2-4\times2\times26$ ✓ $\bullet^5$ ✓

$=64-208<0$ so no real roots $\bullet^6$ ✗

Response 5 : Acceptable communication marks

Method 1

Candidate J

$\sqrt{8^2-4\times2\times26}$ ✓ $\bullet^5$

$=\sqrt{-144}$

not valid

so no real roots ✓ $\bullet^6$

Candidate K

Discriminant $=\sqrt{8^2-4\times2\times26}$ ✓ $\bullet^5$

$=\sqrt{-144}$

can't find root of negative

so no real roots ✓ $\bullet^6$

Candidate L

no real roots if $b^2-4ac<0$

$64-208=-144$ ✓ $\bullet^6$

✓ $\bullet^5$

Method 2

Candidate M

$\dfrac{-(-4)\pm\sqrt{8^2-4\times2\times26}}{2\times2}$ ✓ $\bullet^5$

$=\dfrac{4\pm\sqrt{-144}}{4}$

no $\sqrt{-ve}$

so no real roots ✓ $\bullet^6$

Looking for more resources? Visit scotpapers.com - Scotland's #1 Past Paper Archive

2 (a) Relative to a suitable set of coordinate axes, diagram 1 shows the line $2x-y+5=0$ intersecting the circle $x^2+y^2-6x-2y-30=0$ at the points P and Q.

Find the coordinates of P and Q. **6**

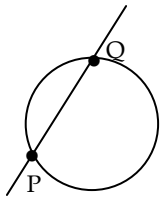


Diagram 1

Generic Scheme

2 (a)

- $\bullet^1$ ss rearrange linear equation
- $\bullet^2$ ss substitute into circle
- $\bullet^3$ pd express in standard form
- $\bullet^4$ pd start to solve
- $\bullet^5$ ic state roots
- $\bullet^6$ pd determine corresponding y-coordinates

Illustrative Scheme

Substituting for y

$\bullet^1$ $y=2x+5$ **stated, or implied by $\bullet^2$**

$\bullet^2$ $\ldots(2x+5)^2\ldots-2(2x+5)\ldots$

$\bullet^3$ $5x^2+10x-15$

$\bullet^4$ e.g. $5(x+3)(x-1)$ } = 0 must appear at the $\bullet^3$ or $\bullet^4$ stage to gain $\bullet^3$.

$\bullet^5$ $x=-3$ and $x=1$

$\bullet^6$ $y=-1$ and $y=7$

Substituting for x

$\bullet^1$ $x=\dfrac{y-5}{2}$ **stated, or implied by $\bullet^2$**

$\bullet^2$ $\left(\dfrac{y-5}{2}\right)^2\ldots-6\left(\dfrac{y-5}{2}\right)\ldots$

$\bullet^3$ $5y^2-30y-35$

$\bullet^4$ e.g. $5(y+1)(y-7)$ } = 0 must appear at the $\bullet^3$ or $\bullet^4$ stage to gain $\bullet^3$.

$\bullet^5$ $y=-1$ and $y=7$

$\bullet^6$ $x=-3$ and $x=1$

6

Notes

1. At $\bullet^4$ the quadratic must lead to two real distinct roots for $\bullet^5$ and $\bullet^6$ to be available.
2. Cross marking is available here for $\bullet^5$ and $\bullet^6$.
3. Candidates do not need to distinguish between points P and Q.

Regularly occurring responses

Response 1 : Solving quadratic equation

Candidate A

✓ $\bullet^1$ ✓ $\bullet^2$

$5x^2+10x+5=0$ ✗ $\bullet^3$

$5(x+1)(x+1)$ ✗ $\bullet^4$

$x=-1$ ✗ $\bullet^5$

$y=3$ ✗ $\bullet^6$

Candidate B

$y=2x+5$ ✓ $\bullet^1$

$x^2+(2x+5)^2-6x-2(7x+5)-30=0$ ✗ $\bullet^2$

$5x^2-15=0$ ✗ $\bullet^3$

$x^2=3$ ✗ $\bullet^4$

$x=\pm\sqrt{3}$ ✗ $\bullet^5$

$y=8\cdot5,\ 1\cdot5$ ✗ $\bullet^6$

Candidate C

✓ $\bullet^1$ ✓ $\bullet^2$

$5x^2+10x-15=0$ ✓ $\bullet^3$

$5x^2+10x=15$

$5x(x+2)=15$ ✗ $\bullet^4$

$x(x+2)=3$

$x=3$ $x=1$ ✗ $\bullet^5$

$y=11$ $y=7$ ✗ $\bullet^6$

Cross marking is **not** available here for $\bullet^5$ and $\bullet^6$, as there are no distinct roots. See Note 1.

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

2 (b) Diagram 2 shows the circle from (a) and a second congruent circle, which also passes through P and Q.

Determine the equation of this second circle. **6**

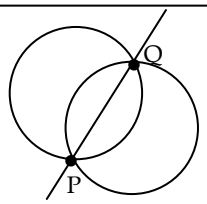


Diagram 2

Generic Scheme

2 (b)

$\bullet^{7}$ ic centre of original circle

$\bullet^{8}$ pd radius of original circle

Method 1 : Using midpoint

$\bullet^{9}$ ss midpoint of chord

$\bullet^{10}$ ss evidence for finding new centre

$\bullet^{11}$ ic centre of new circle

$\bullet^{12}$ ic equation of new circle

Method 2 : Stepping out using P and Q

$\bullet^{9}$ ss evidence of C_1 to P **or** C_1 to Q

$\bullet^{10}$ ss evidence of Q to C_2 **or** P to C_2

$\bullet^{11}$ ic centre of new circle

$\bullet^{12}$ ic equation of new circle

Illustrative Scheme

$\bullet^{7}$ (3, 1)

$\bullet^{8}$ $\sqrt{40}$ Accept $r^2 = 40$

Method 1 : Using midpoint

$\bullet^{9}$ (–1, 3)

$\bullet^{10}$ e.g. stepping out or midpoint formula

$\bullet^{11}$ (–5, 5)

$\bullet^{12}$ $(x+5)^2+(y-5)^2=40$

Method 2 : Stepping out using P and Q

$\bullet^{9}$ e.g. stepping out or vector approach

$\bullet^{10}$ e.g. stepping out or vector approach

$\bullet^{11}$ (–5, 5)

$\bullet^{12}$ $(x+5)^2+(y-5)^2=40$ **6**

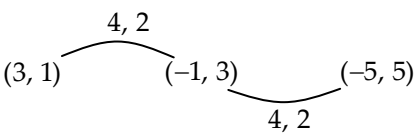
Notes

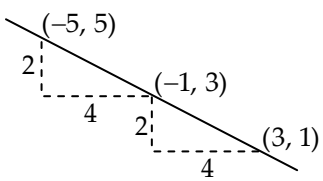
4. The evidence for $\bullet^{7}$ and $\bullet^{8}$ may appear in (a).
5. Centre (–5, 5) **without working** in method 1 may still gain $\bullet^{12}$ but not $\bullet^{10}$ or $\bullet^{11}$, in method 2 may still gain $\bullet^{12}$ but not $\bullet^{9}$, $\bullet^{10}$ or $\bullet^{11}$.

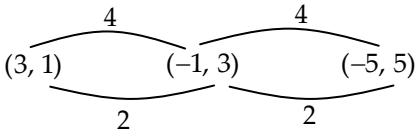
   Any other centre **without working** in method 1 does not gain $\bullet^{10}$, $\bullet^{11}$ or $\bullet^{12}$, in method 2 does not gain $\bullet^{9}$, $\bullet^{10}$, $\bullet^{11}$ or $\bullet^{12}$.
6. The centre must have been clearly indicated before it is used at the $\bullet^{12}$ stage.
7. Do not accept e.g. $\sqrt{40}^2$ or 39·69, or any other decimal approximations for $\bullet^{12}$.
8. The evidence for $\bullet^{8}$ may not appear until the candidate states the radius or equation of the second circle.

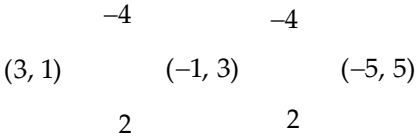
Regularly occurring responses

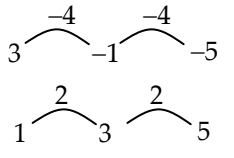
Response 2 : Examples of evidence for stepping out for $\bullet^{10}$ in method 1 or $\bullet^{9}$ or $\bullet^{10}$ in method 2



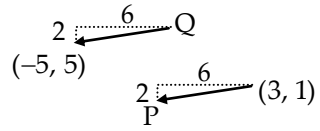




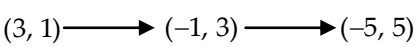


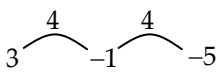


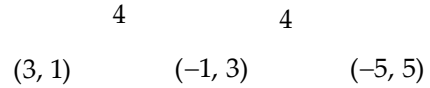
For method 2



Response 3 : Examples of evidence which do not gain $\bullet^{10}$ in method 1 for stepping out







Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

3 A function f is defined on the domain $0 \le x \le 3$ by $f(x) = x^3 - 2x^2 - 4x + 6$.

Determine the maximum and minimum values of f. **7**

Generic Scheme

3

- $\bullet^1$ ss start to differentiate
- $\bullet^2$ ss complete derivative and set to 0
- $\bullet^3$ pd start to solve $f'(x) = 0$
- $\bullet^4$ pd solve $f'(x) = 0$
- $\bullet^5$ ic evaluate f at relevant stationary point
- $\bullet^6$ ss consider end-points
- $\bullet^7$ ic state max. and min. values

Illustrative Scheme

- $\bullet^1$ differentiate x^3 or $-2x^2$ correctly
- $\bullet^2$ $3x^2 - 4x - 4$
- $\bullet^3$ e.g. $(3x+2)(x-2)$

} $= 0$ must appear at $\bullet^2$ or $\bullet^3$ to gain $\bullet^2$.

- $\bullet^4$ $-\frac{2}{3}$, 2
- $\bullet^5$ $f(2) = -2$
- $\bullet^6$ $f(0) = 6$ **and** $f(3) = 3$
- $\bullet^7$ max. 6 **and** min. -2

7

Notes

1. The only valid approach is via differentiation. A numerical approach can only gain $\bullet^6$.
2. Candidates who consider stationary points only cannot gain $\bullet^6$ or $\bullet^7$.
3. Treat maximum (0, 6) and minimum (2, −2) as bad form.
4. Cross marking is **not** applicable to $\bullet^6$ or $\bullet^7$.
5. Ignore any nature table which may appear in a candidate's solution, however (2, −2) at table is sufficient for $\bullet^5$.

Regularly occurring responses

Response 1 : Algebraic issues in working

Candidate A

$y' = 3x^2 - 4x - 4$ ✓ $\bullet^1$ ✓
$(3x-2)(x+2)$ ✗ $\bullet^2$ ✗
$x = \frac{2}{3}$, $x = -2$ ✓✗ $\bullet^3$ ✗
When $x = \frac{2}{3}$, $y = \frac{74}{27}$ ✓✗ $\bullet^4$ ✗
$f(0) = 6$ and $f(3) = 3$ ✓ $\bullet^5$ ✗
max = 6, min $= 2\frac{20}{27}$ ✓✗ $\bullet^6$ ✓
$\bullet^7$ ✗

Candidate B

$3x^2 - 4x - 4 = 0$ ✓ $\bullet^1$ ✓
$(3x-2)(x-2)$ ✗ $\bullet^2$ ✓
$x = \frac{2}{3}$ or $x = 2$ ✓✗ $\bullet^3$ ✗
so $f(2) = -2$ ✓✗ $\bullet^4$ ✗
$\bullet^5$ ✓✗

Since $\frac{2}{3}$ is within the domain, $f\left(\frac{2}{3}\right)$ must also be calculated to gain $\bullet^5$.

Candidate C

$3x^2 - 4x - 4$ ✓
$(3x+2)(x-2)$ ✓ $\bullet^1$ ✓
$3x+2=0 \quad x-2=0$ $\bullet^2$ ✗
$x = -\frac{2}{3} \quad x = 2$ ✓ $\bullet^3$ ✓
$f(2) = -2$ ✓ $\bullet^4$ ✓
$\bullet^5$ ✓

Ignore the value of $f\left(-\frac{2}{3}\right)$ here, if it is included.

Response 2 : Derivative not explicitly set to zero

Candidate D

$f'(x) = 3x^2 - 4x - 4$ ✓ $\bullet^1$
$f'(x) = 0$ ✓ $\bullet^2$

Candidate E

$f'(x) = 0$
$f'(x) = 3x^2 - 4x - 4$ ✓ $\bullet^1$ ✓ $\bullet^2$
$= (3x+2)(x-2)$ ✓ $\bullet^3$

Candidate F

$f'(x) = 0$
$3x^2 - 4x - 4$ ✓ $\bullet^1$ ✗ $\bullet^2$
$= (3x+2)(x-2)$ ✓ $\bullet^3$

Candidate G

$f'(x) = 0$ only ✗ $\bullet^1$ ✗ $\bullet^2$

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

3 continued

Regularly occurring responses

Response 3 : Solving quadratic equation

Candidate H

$f'(x) = 3x^2 - 4x - 4$ ✓ $\bullet^1$ ✓

$3x^2 - 4x - 4 = 0$ ✓ $\bullet^2$ ✓

$3x^2 - 4x = 4$ $\bullet^3$ ✗

$x(3x - 4) = 4$ $\bullet^4$ ✗

$x = 4,\ \frac{4}{3}$ $\bullet^5$ ✗

Candidate I

✓ $\bullet^1$

$3x^2 - 4x - 4 = 0$ ✓ $\bullet^2$

$x = \dfrac{-(-4) \pm \sqrt{(4)^2 - 4 \times 3 \times (-4)}}{2 \times 3}$ ✓ $\bullet^3$

Ignore omission of negative sign at square here.

Due to 'method' chosen $\bullet^3$, $\bullet^4$, $\bullet^5$ and $\bullet^7$ are not available.

Response 4 : Numerical approach

Candidate J

$f(0) = 6$

$f(3) = 3$ ✓ $\bullet^6$

This candidate has stayed within the interval $0 \le x \le 3$.

Candidate K

$f(0) = 6$

$f(1) = 1$

$f(2) = -2$ ✗ $\bullet^5$

$f(3) = 3$ ✓ $\bullet^6$

Candidate L

$f(0) = 6$

$f(1) = 1$

$f(2) = -2$ ✗ $\bullet^5$

$f(3) = 3$ ✗ $\bullet^6$

$f(4) = 22$

This candidate has gone outwith the interval $0 \le x \le 3$.

For $\bullet^5$, $f(2)$ must come from calculus and not from any other approach.

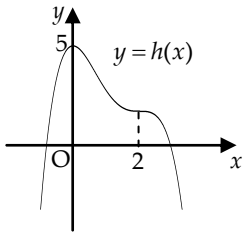
Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

4 The diagram below shows the graph of a quartic $y = h(x)$, with stationary points at $x = 0$ and $x = 2$.

On separate diagrams sketch the graphs of:

(a) $y = h'(x)$; **3**

(b) $y = 2 - h'(x)$. **3**



Generic Scheme

4 (a)

- $\bullet^1$ ic identify roots
- $\bullet^2$ ic interpret point of inflection
- $\bullet^3$ ic complete cubic curve

- $\bullet^1$ 0 **and** 2 only
- $\bullet^2$ turning point at (2, 0)
- $\bullet^3$ cubic, passing through O with negative gradient

3

Notes

1. All graphs must include both the x and y axes (labelled or unlabelled), however the origin need not be labelled.
2. No marks are available unless a graph is attempted.
3. No marks are available to a candidate who makes several attempts at a graph on the same diagram, unless it is clear which is the final graph.
4. A linear graph gains no marks in both (a) and (b).

4 (b)

- $\bullet^4$ ic reflection in x-axis
- $\bullet^5$ ic translation $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$
- $\bullet^6$ ic annotation of 'transformed' graph

- $\bullet^4$ reflection of graph in (a) in x-axis
- $\bullet^5$ graph moves parallel to y-axis by 2 units upwards
- $\bullet^6$ two 'transformed' points appropriately annotated (see Note 5)

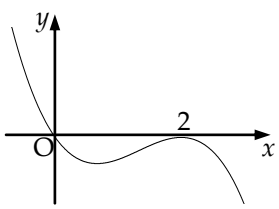
3

Notes

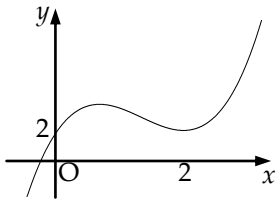
5. 'Transformed' here means a reflection followed by a translation.
6. $\bullet^4$ and $\bullet^5$ apply to the entire curve.
7. In each of the following circumstances :
   - Candidates who transform the original graph
   - Candidates who sketch a parabola in (a)

   mark the candidate's attempt as normal and unless a mark of 0 has been scored, deduct the last mark awarded. Indicate this with ✗ (see Regular occurring response G).
8. A reflection in any line parallel to the y-axis does not gain $\bullet^4$ or $\bullet^6$.
9. A translation other than $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$ does not gain $\bullet^5$ or $\bullet^6$.

Graph for (a)



Graph for (b)



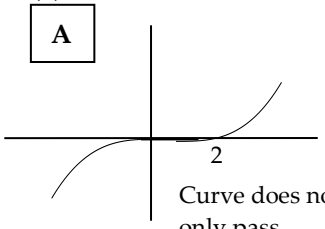
- $\bullet^4$ ic reflection in $x-$axis

4 continued

Regularly occurring responses

In (a)

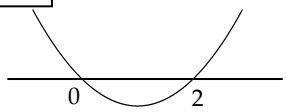
A



Curve does not only pass through 0 and 2 on x-axis.

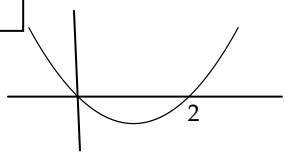
✗ •1
✗ •2
✗ •3

B



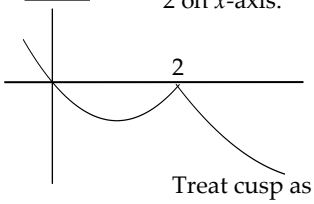
✗ •1
✗ •2
✗ •3

C



✓ •1
✗ •2
✗ •3

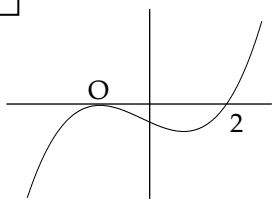
D



Treat cusp as turning point.

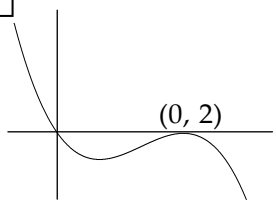
✓ •1
✓ •2
✓ •3

E



✗ •1
✗ •2
✗ •3

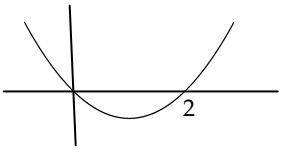
F



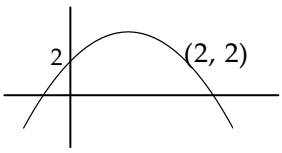
✓ •1
✓ •2
✓ •3

In (a) and (b)

G



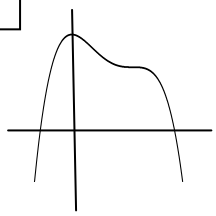
✓ •1
✗ •2
✗ •3



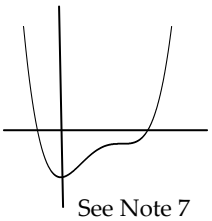
See Note 7

✗ •4
✗ •5
✗ •6

H



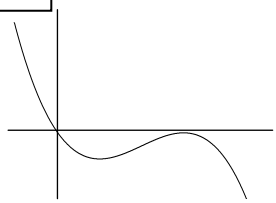
✗ •1
✗ •2
✗ •3



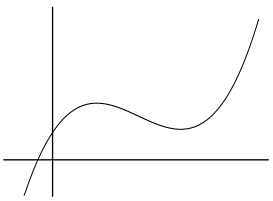
See Note 7

✗ •4
✗ •5
✗ •6

I

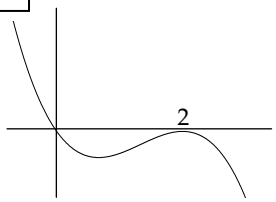


✗ •1
✗ •2
✓ •3

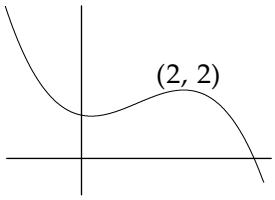


✓ •4
✗ •5
✗ •6

J



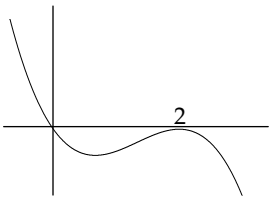
✓ •1
✓ •2
✓ •3



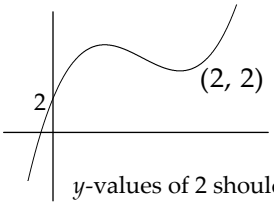
See Note 6

✗ •4
✗ •5
✗ •6

K



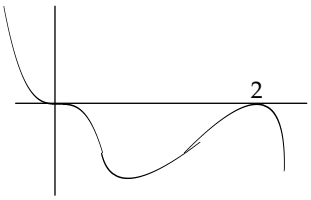
✓ •1
✓ •2
✓ •3



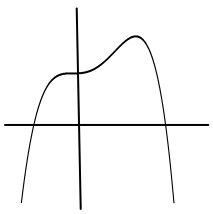
y-values of 2 should be roughly in line – too far apart here

✓ •4
✓ •5
✗ •6

L



✓ •1
✓ •2
✗ •3



✗ •4
✗ •5
✗ •6

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

5 A is the point $(3, -3, 0)$, B is $(2, -3, 1)$ and C is $(4, k, 0)$.

(a) (i) Express $\overrightarrow{BA}$ and $\overrightarrow{BC}$ in component form.

(ii) Show that $\cos A\hat{B}C = \dfrac{3}{\sqrt{2(k^2+6k+14)}}$. **7**

Generic Scheme

5(a)

- $\bullet^1$ ic interpret vector
- $\bullet^2$ pd process vector
- $\bullet^3$ ss use scalar product
- $\bullet^4$ pd find scalar product
- $\bullet^5$ pd find $\left|\overrightarrow{BA}\right|$
- $\bullet^6$ ic find expression for $\left|\overrightarrow{BC}\right|$
- $\bullet^7$ ic complete to result

Illustrative Scheme

- $\bullet^1$ $\begin{pmatrix}1\\0\\-1\end{pmatrix}$ Treat $\begin{pmatrix}1\\0\\-1\end{pmatrix}$ written as $(1, 0, -1)$ as bad form.
- $\bullet^2$ $\begin{pmatrix}2\\k+3\\-1\end{pmatrix}$
- $\bullet^3$ $\cos A\hat{B}C = \dfrac{\overrightarrow{BA}\,.\,\overrightarrow{BC}}{\left|\overrightarrow{BA}\right|\left|\overrightarrow{BC}\right|}$ see Note 1
- $\bullet^4$ 3
- $\bullet^5$ $\sqrt{2}$
- $\bullet^6$ $\sqrt{2^2+(k+3)^2+(-1)^2}$ or equivalent
- $\bullet^7$ $\dfrac{3}{\sqrt{2}\sqrt{k^2+6k+14}}$ **and** $\dfrac{3}{\sqrt{2(k^2+6k+14)}}$

  or $\left|\overrightarrow{BA}\right|\left|\overrightarrow{BC}\right| = \sqrt{2}\times\sqrt{k^2+6k+14}$ **and** $\dfrac{3}{\sqrt{2(k^2+6k+14)}}$

7

Notes

1. If the evidence for $\bullet^3$ does not appear explicitly, then $\bullet^3$ is only awarded if working for $\bullet^7$ is attempted.
2. $\bullet^7$ is dependent on gaining $\bullet^4$, $\bullet^5$ and $\bullet^6$.

Regularly occurring responses

Response 1 : Calculating wrong angle

Candidate A

$\cos AOC = \dfrac{\overrightarrow{OA}.\overrightarrow{OC}}{\left|\overrightarrow{OA}\right|\left|\overrightarrow{OC}\right|}$ ✗ $\bullet^3$

$\overrightarrow{OA}.\overrightarrow{OC} = 3\times4+(-3)\times k+0\times0 = 12-3k$ ✓ $\bullet^4$

$\left|\overrightarrow{OA}\right| = \sqrt{18}$ ✓ $\bullet^5$

$\left|\overrightarrow{OC}\right| = \sqrt{16+k^2}$ ✓ $\bullet^6$

$\cos ABC = \dfrac{12-3k}{\sqrt{18}\sqrt{16+k^2}}$ ✗ $\bullet^7$

Candidate B

$\cos AOB = \dfrac{\overrightarrow{OA}.\overrightarrow{OB}}{\left|\overrightarrow{OA}\right|\left|\overrightarrow{OB}\right|}$ ✗ $\bullet^3$

$\overrightarrow{OA}.\overrightarrow{OB} = 3\times2+(-3)\times(-3)+0\times1 = 15$ ✓ $\bullet^4$

$\left|\overrightarrow{OA}\right| = \sqrt{18}$ ✓ $\bullet^5$

$\left|\overrightarrow{OB}\right| = \sqrt{14}$ ✗ $\bullet^6$

$\cos ABC = \dfrac{15}{\sqrt{18}\sqrt{14}}$ ✗ $\bullet^7$

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

5 (b) If angle ABC = 30°, find the possible values of k. **5**

Generic Scheme

5(b)

Method 1 : Squaring first

- $\bullet^8$ ic link with (a)
- $\bullet^9$ ss square both sides
- $\bullet^{10}$ pd rearrange into 'non-fractional' format
- $\bullet^{11}$ pd write in standard form
- $\bullet^{12}$ pd solve for k

Method 2 : Dealing with fractions first

- $\bullet^8$ ic link with (a)
- $\bullet^9$ pd rearrange into 'non-fractional' format
- $\bullet^{10}$ ss square both sides
- $\bullet^{11}$ pd write in standard form
- $\bullet^{12}$ pd solve for k

Illustrative Scheme

Method 1 : Squaring first

- $\bullet^8$ $\dfrac{3}{\sqrt{2(k^2+6k+14)}} = \cos 30^\circ$
- $\bullet^9$ $\left(\dfrac{3}{\sqrt{2(k^2+6k+14)}}\right)^2 = \left(\dfrac{\sqrt{3}}{2}\right)^2$
- $\bullet^{10}$ $k^2+6k+14=6$ or equivalent
- $\bullet^{11}$ $k^2+6k+8=0$ or equivalent

= 0 must appear at this stage.

- $\bullet^{12}$ $k=-2$ or -4

Method 2 : Dealing with fractions first

- $\bullet^8$ $\dfrac{3}{\sqrt{2(k^2+6k+14)}} = \cos 30^\circ$
- $\bullet^9$ $\sqrt{3}\sqrt{2(k^2+6k+14)} = 6$
- $\bullet^{10}$ $6(k^2+6k+14)=36$
- $\bullet^{11}$ $k^2+6k+8=0$ or equivalent

= 0 must appear at this stage.

- $\bullet^{12}$ $k=-2$ or -4

5

Notes

3. The evidence for $\bullet^9$ may appear in the working for $\bullet^{10}$ in both methods.
4. $\bullet^9$ is the only mark available to candidates who replace cos 30° by 30 in method 1 and $\bullet^{10}$ in method 2.
5. All 5 marks are available to candidates who use 0·87 for cos 30° but 0·9 can gain a maximum of 4 marks.

Regularly occurring responses

Response 2 : Working with cos 30° throughout the question

Candidate C (Method 1)

$\cos 30^\circ = \dfrac{3}{\sqrt{2(k^2+6k+14)}}$ ✓ $\bullet^8$

$(\cos 30^\circ)^2 = \left(\dfrac{3}{\sqrt{2(k^2+6k+14)}}\right)^2$ ✓ $\bullet^9$

$(\cos 30^\circ)^2 = \dfrac{9}{2(k^2+6k+14)}$

$2(\cos 30^\circ)^2(k^2+6k+14) = 9$ ✓ $\bullet^{10}$

If cos 30° is subsequently evaluated then $\bullet^{11}$ and $\bullet^{12}$ may still be available.

Response 3 : Using the wrong value for cos 30°

Candidate D (Method 2)

$\dfrac{3}{\sqrt{2(k^2+6k+14)}} = \dfrac{1}{2}$ ✗ $\bullet^8$

$\sqrt{2(k^2+6k+14)} = 6$ ✗ $\bullet^9$

$2(k^2+6k+14) = 36$ ✗ $\bullet^{10}$

$k^2+6k+14 = 18$

$k^2+6k-4=0$ ✗ $\bullet^{11}$

$k = \dfrac{-6 \pm \sqrt{6^2-4\times1\times(-4)}}{2\times1}$

$= 0{\cdot}61, -6{\cdot}61$ ✗ $\bullet^{12}$

6 For $0 < x < \frac{\pi}{2}$, sequences can be generated using the recurrence relation

$$u_{n+1} = (\sin x)u_n + \cos 2x, \text{ with } u_0 = 1.$$

(a) Why do these sequences have a limit? **2**

Generic Scheme **Illustrative Scheme**

6 (a)

$\bullet^1$ ic condition on u_n coefficient	$\bullet^1$ $-1 < \sin x < 1$
$\bullet^2$ ic connect coefficient with given interval	$\bullet^2$ in interval, $0 < \sin x < 1$

2

Notes

1. For $\bullet^1$ **do not** accept:
   - $\sin x$ lies between -1 and 1
   - $-1 < x < 1$
   - $-1 < \sin < 1$

   However, accept '$\sin x$ greater than -1 and less than 1' for $\bullet^1$.
2. Do not accept $-1 < a < 1$ for $\bullet^1$ unless a is clearly identified as $\sin x$, which may not appear until (b).
3. $0 < \sin x < 1$ and nothing else, does not gain $\bullet^1$ but gains $\bullet^2$.
4. $0 \le \sin x \le 1$ and nothing else, does not gain $\bullet^1$ or $\bullet^2$.

Regularly occurring responses

Response 1 : Attempts at giving a reason for limit

Candidate A

This sequence has a limit because $-1 < a < 1$, ✓ $-1 < \sin x < 1$ within the domain. ^

$\bullet^1$ ✓ $\bullet^2$ X

Candidate B

^ Since $\sin x$ in this domain will always be greater than 0 and less than 1. ✓

$\bullet^1$ X $\bullet^2$ ✓

Candidate C

$\sin \frac{\pi}{2} = 1$ and $\sin 0 = 0$ so the multiplier of u_n is between 0 and 1, so it has a limit. ^ ^

$\bullet^1$ X $\bullet^2$ X

Candidate D

$-1 \le \sin x \le 1$,

for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ ✓

so limit exists ^

$\bullet^1$ X $\bullet^2$ ✓

Response 2 : Minimum response for both marks

Candidate E

for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ $\bullet^2$ ✓

so $-1 < \sin x < 1$ $\bullet^1$ ✓

so limit

Candidate F

if limit, $-1 < \sin x < 1$ $\bullet^1$ ✓

for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ $\bullet^2$ ✓

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

6 (b) The limit of one particular sequence generated by this recurrence relation is $\frac{1}{2}\sin x$.

Find the value(s) of x. **7**

Generic Scheme

6 (b)

- $\bullet^3$ ss appropriate limit method
- $\bullet^4$ ic substitute for limit
- $\bullet^5$ ss use appropriate double angle formula
- $\bullet^6$ pd express in standard form
- $\bullet^7$ pd start to solve quadratic equation
- $\bullet^8$ pd reduce to equations in $\sin x$ only
- $\bullet^9$ ic select valid solution

Illustrative Scheme

- $\bullet^3$ limit $= \dfrac{\cos 2x}{1-\sin x}$ **or** $l = \sin x \times l + \cos 2x$
- $\bullet^4$ $\dfrac{1}{2}\sin x = \dfrac{\cos 2x}{1-\sin x}$ **or** $\dfrac{1}{2}\sin x = \sin x \times \dfrac{1}{2}\sin x + \cos 2x$

  ($\bullet^3$ **may be stated, or implied by** $\bullet^4$ in both methods)
- $\bullet^5$ $\ldots 1 - 2\sin^2 x \ldots$
- $\bullet^6$ e.g. $3\sin^2 x + \sin x - 2$
- $\bullet^7$ e.g. $(3\sin x - 2)(\sin x + 1)$

  } $= 0$ must appear at $\bullet^6$ or $\bullet^7$ to gain $\bullet^6$.
- $\bullet^8$ $\sin x = \frac{2}{3}$ or $\sin x = -1$
- $\bullet^9$ $x = 0{\cdot}730$ or outwith interval

7

Notes

5. $\bullet^7$, $\bullet^8$ and $\bullet^9$ are only available if a quadratic equation is obtained at $\bullet^6$ stage.
6. Candidates may express the quadratic equation at the $\bullet^6$ stage in the form $3s^2 + s - 2 = 0$. For candidates who do not solve a trigonometric quadratic equation at $\bullet^7$ $\sin x$ must appear explicitly to gain $\bullet^8$.
7. $\bullet^7$, $\bullet^8$ and $\bullet^9$ are not available to candidates who 'solve' a quadratic equation in the form $ax^2 + bx = c,\ c \neq 0$.
8. For $\bullet^9$ there must be one valid solution, and one solution outwith interval which is rejected.
9. $\bullet^9$ is not available to candidates who leave their answer in degree measure.
10. Cross marking is available for $\bullet^8$ and $\bullet^9$.

Regularly occurring responses

Response 3 : Evidence for identification of a appearing in (b)

Candidate G

(a) $-1 < a < 1$ $\bullet^1$ ✓

(b) $L = \dfrac{b}{1-a} = \dfrac{\cos 2x}{1-\sin x}$ ✓ $\bullet^3$ ✓ $\bullet^1$

$\bullet^2$ ✗

$\bullet^3$ ✓

Response 4 : Error in algebra and subsequent quadratic equation solution

Candidate H

$L = \dfrac{b}{1-a} = \dfrac{1}{2}\sin x$

$\dfrac{\cos 2x}{1-\sin x} = \dfrac{1}{2}\sin x$ ✓ $\bullet^3$ ✓ $\bullet^4$

$\cos 2x = -\frac{1}{2}\sin^2 x$ ✗ $\bullet^6$

$\frac{1}{2}\sin^2 x + \cos 2x = 0$

$\frac{1}{2}\sin^2 x + (1 - 2\sin^2 x) = 0$ ✗ $\bullet^5$

$-\frac{3}{2}\sin^2 x + 1 = 0$

$\sin^2 x = \frac{2}{3}$ ✗ $\bullet^7$

$\sin x = \sqrt{\frac{2}{3}}$ and $\sin x = -\sqrt{\frac{2}{3}}$ ✗ $\bullet^8$

$x = 0{\cdot}955,\ 2{\cdot}186$ $\quad x = 4{\cdot}097,\ 5{\cdot}328$ ✗ $\bullet^9$

Candidate I

$\dfrac{\cos 2x}{1-\sin x} = \dfrac{1}{2}\sin x$ ✓ $\bullet^3$ ✓ $\bullet^4$

$\frac{1}{2}\sin x(1-\sin x) = 1 - \sin^2 x$ ✗ $\bullet^5$

$\sin^2 x + \sin x - 2 = 0$ ✗ $\bullet^6$

$(\sin x - 1)(\sin x + 2) = 0$ ✗ $\bullet^7$

$\sin x = 1$ and $\sin x = -2$ ✗ $\bullet^8$

$x = \frac{\pi}{2}$ not possible ✗ $\bullet^9$

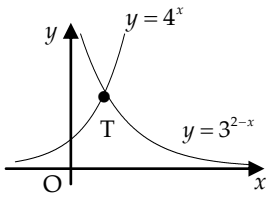
See Note 8

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

7 The diagram shows the curves with equations $y = 4^x$ and $y = 3^{2-x}$.

The graphs intersect at the point T.

(a) Show that the x-coordinate of T can be written in the form $\dfrac{\log_a p}{\log_a q}$, for all $a > 1$.



6

Generic Scheme

7(a)

- $\bullet^1$ ss equate expressions for y
- $\bullet^2$ ss take logarithms of both sides
- $\bullet^3$ ic use law of logs : $\log_a x^n = n\log_a x$
- $\bullet^4$ pd gather like terms
- $\bullet^5$ ic use law of logs : $\log_a p + \log_a q = \log_a pq$
- $\bullet^6$ ic complete to required form

In methods 1 and 2:

If the first line of working is that at the $\bullet^2$ stage, then $\bullet^1$ and $\bullet^2$ are awarded.

If the first line of working is that at the $\bullet^3$ stage, then only $\bullet^2$ and $\bullet^3$ are awarded.

Illustrative Scheme

Method 1

- $\bullet^1$ $4^x = 3^{2-x}$
- $\bullet^2$ $\log_a\left(4^x\right) = \log_a\left(3^{2-x}\right)$ **stated, or implied by $\bullet^3$**
- $\bullet^3$ $x\log_a 4 = (2-x)\log_a 3$
- $\bullet^4$ $x(\log_a 4 + \log_a 3) = 2\log_a 3$
- $\bullet^5$ $x\log_a 12 = \log_a 9$
- $\bullet^6$ $\dfrac{\log_a 9}{\log_a 12}$ **stated explicitly**

Method 2

- $\bullet^1$ $4^x = 3^{2-x}$
- $\bullet^2$ $\log_3\left(4^x\right) = 2-x$
- $\bullet^3$ $x\log_3 4 = 2-x$
- $\bullet^4$ $x = \dfrac{2}{1+\log_3 4}$
- $\bullet^5$ $\dfrac{2\log_3 3}{\log_3 12}$
- $\bullet^6$ $\dfrac{\log_a 9}{\log_a 12}$ **stated explicitly**

Method 3

- $\bullet^1$ $4^x = 3^{2-x}$
- $\bullet^2$ $4^x = \dfrac{3^2}{3^x}$
- $\bullet^3$ $12^x = 9$
- $\bullet^4$ $\log_a 12^x = \log_a 9$
- $\bullet^5$ $x\log_a 12 = \log_a 9$
- $\bullet^6$ $\dfrac{\log_a 9}{\log_a 12}$ **stated explicitly**

6

Notes

1. In methods 1 and 2, if no base is indicated then $\bullet^2$ is not available , however $\bullet^3$, $\bullet^4$ and $\bullet^5$ are still available.

   In method 3, if no base is indicated then $\bullet^4$ is not available , however $\bullet^5$ is still available.
2. In all methods, if a numerical base is used then $\bullet^6$ is not available.
3. In method 1, the omission of brackets at the $\bullet^3$ stage is treated as bad form, see Response 1.
4. p and q must be numerical values.

Regularly occurring responses

Response 1 : Omission of brackets around $2-x$

Candidate A

$4^x = 3^{2-x}$ ✓ $\bullet^1$

$x\log_a 4 = 2 - x\log_a 3$ ✓ $\bullet^2$ ✓ $\bullet^3$

Candidate B

$4^x = 3^{2-x}$ ✓ $\bullet^1$

$x\log_a 4 = 2 - x\log_a 3$ ✓ $\bullet^2$ ✓ $\bullet^3$

$x(\log_a 4 + \log_a 3) = 2$ ✗ $\bullet^4$

$x\log_a 12 = 2$ ✗ $\bullet^5$

Response 2 : Using different bases

Candidate C

$4^x = 3^{2-x}$ ✓ $\bullet^1$

$\log_3 4^x = \log_4 3^{2-x}$ ✗ $\bullet^2$

$x\log_3 4 = (2-x)\log_4 3$ ✗ $\bullet^3$

Response 3 : Taking logs first

Candidate D

$y = 4^x$ and $y = 3^{2-x}$

$\log_a y = \log_a 4^x$ and $\log_a y = \log_a 3^{2-x}$ ✓ $\bullet^2$

$\log_a y = x\log_a 4$ and $\log_a y = (2-x)\log_a 3$ ✓ $\bullet^3$

$x\log_a 4 = (2-x)\log_a 3$ ✓ $\bullet^1$

$x = \dfrac{2}{\log_a 12}$

$= \dfrac{2\log_a a}{\log_a 12}$

$= \dfrac{\log_a a^2}{\log_a 12}$ ✗ $\bullet^6$

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

7 (b) Calculate the y-coordinate of T.	**2**

Generic Scheme **Illustrative Scheme**

7(b)

Generic Scheme	Illustrative Scheme
$\bullet^7$ ic substitute in for x	$\bullet^7$ e.g. $y = 4^{\frac{\log_a 9}{\log_a 12}}$ **stated, or implied by** $\bullet^8$
$\bullet^8$ pd process y	$\bullet^8$ e.g. $y \approx 4^{0\cdot8842} \approx 3\cdot4$

2

Notes

5. Candidates must work to at least two significant figures in (b) e.g. $4^{0\cdot9} = 3\cdot5$ does not gain $\bullet^8$, but $\bullet^7$ is available.
6. $\bullet^8$ is only available if the power used comes from $\dfrac{\log_a p}{\log_a q}$ in (a).

Regularly occurring responses

Response 4 : Using p and q as integer values without working

Candidate E

$\left.\begin{matrix} p=4 \\ q=3 \end{matrix}\right\} y = 4^{1\cdot26} = 5\cdot74 \text{ or } 5\cdot75$ X $\bullet^7$ X $\bullet^8$

Candidate F

$\left.\begin{matrix} p=3 \\ q=4 \end{matrix}\right\} y = 4^{0\cdot79} = 2\cdot99 \text{ or } 3$ X $\bullet^7$ X $\bullet^8$

Response 5 : Using integer values calculated in (a)

Candidate G

$\left.\begin{matrix} p=10 \\ q=4 \end{matrix}\right\} y = 4^{2\cdot5} = 32$ X $\bullet^7$ X $\bullet^8$

[END OF MARKING INSTRUCTIONS]

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive